

Off-Policy KTM Pipeline: End-to-End Methodology

Geometric Flow Repository

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1 Goal

This document describes the full experimental pipeline implemented in:

- `offpolicy_ktm_pipeline/run_full_pipeline.py`
- `offpolicy_ktm_pipeline/src/*.py`

The pipeline has two main parts:

1. Build a contextual logging dataset (θ, δ, r, o) from student-item logs.
2. Learn and evaluate a new policy for item difficulty selection with off-policy estimators.

2 Data Processing and KTM Latents

2.1 PIX preprocessing

For PIX-like data, `data/pix_processing.py` applies:

- ID remapping to contiguous integers for user/item/skill.
- `answer_result` mapping:

$$\text{ok} \mapsto 1, \quad \text{ko} \mapsto 0, \quad \text{aband} \mapsto 0.5$$

- Binary outcome column:

$$\text{outcome} = \begin{cases} 0 & \text{if } \text{answer_result} = 0.5 \\ \text{answer_result} & \text{otherwise} \end{cases}$$

2.2 KTM-style latent construction

`models/ktm.py::processing_data` fits a one-hot user-item logistic model:

$$\Pr(y = 1 \mid u, i) = \sigma(\beta_u + \gamma_i + b)$$

Then defines:

$$\theta_u = \beta_u, \quad \delta_i = -\gamma_i$$

where θ_u is proficiency and δ_i is item difficulty.

It then:

- ranks users and items by latent value (for stable integer IDs),
- attaches **proficiency** and **difficulties**,
- builds sequence fields:

`sequence_position, sequence_length, order_sequence.`

3 Round Construction

The pipeline trains/evaluates by rounds indexed by `order_sequence`. If needed, long tails are compressed to a fixed number of rounds (`--max-rounds`) with:

- **tail**: keep early rounds, merge tail into final bucket.
- **quantile**: weighted quantile bucketing by observation count.
- **tail_count_at_n**: after round N , aggregate tail windows to roughly match count at round N .

4 Behavior Policy via Gaussian Regression

For each round o , `fit_gaussian_by_order` fits:

$$\delta \mid \theta, o \sim \mathcal{N}(\mu_o(\theta), \sigma_o(\theta)^2)$$

with polynomial parameterization:

$$\mu_o(\theta) = \sum_{d=0}^{D_\mu} \beta_{o,d}^{(\mu)} \theta^d, \quad \log \sigma_o(\theta) = \sum_{d=0}^{D_\sigma} \beta_{o,d}^{(\sigma)} \theta^d$$

and hard floor:

$$\sigma_o(\theta) = \max(\exp(\log \sigma_o(\theta)), \sigma_{\min}),$$

where `sigma_floor` is typically 0.2.

The per-round fit minimizes average Gaussian negative log-likelihood:

$$\mathcal{L}_{\text{NLL}} = \frac{1}{n} \sum_{j=1}^n \left[\frac{1}{2} \log(2\pi) + \log \sigma_o(\theta_j) + \frac{1}{2} \left(\frac{\delta_j - \mu_o(\theta_j)}{\sigma_o(\theta_j)} \right)^2 \right].$$

5 Propensity and Reward

After attaching round-specific behavior parameters, each tuple gets:

$$\log \pi_b(\delta \mid \theta, o) = -\frac{1}{2} \log(2\pi) - \log \sigma_o(\theta) - \frac{1}{2} \left(\frac{\delta - \mu_o(\theta)}{\sigma_o(\theta)} \right)^2.$$

Reward is defined as:

$$r = \text{correct} \cdot (\delta - \delta_{\min}),$$

so successful responses on harder items get larger reward.

6 Learned Policy and Training Objective

6.1 Policy class

`models/offpolicy_gaussian_policy.py::GlobalGaussianPolicy` learns one global policy:

$$\pi_\phi(\delta \mid \theta) = \mathcal{N}(\mu_\phi(\theta), \sigma_\phi(\theta)^2),$$

$$\mu_\phi(\theta) = \sum_{d=0}^{D_\mu} \alpha_d \theta^d, \quad \log \sigma_\phi(\theta) = \sum_{d=0}^{D_\sigma} \eta_d \theta^d, \quad \sigma_\phi(\theta) \geq \sigma_{\min}.$$

6.2 Importance weights

For each sample:

$$w = \exp(\log \pi_\phi(\delta \mid \theta) - \log \pi_b(\delta \mid \theta, o)), \quad \tilde{w} = \min(w, c_{\text{clip}}).$$

The code also clips log-ratio and max weight for stability.

6.3 Optimized objectives

Supported objectives (`--objective`):

$$\hat{V}_{\text{IPS}} = \frac{1}{n} \sum_i w_i r_i, \quad \hat{V}_{\text{SNIPS}} = \frac{\sum_i w_i r_i}{\sum_i w_i},$$

$$\hat{V}_{\text{CIPS}} = \frac{1}{n} \sum_i \tilde{w}_i r_i, \quad \hat{V}_{\text{CSNIPS}} = \frac{\sum_i \tilde{w}_i r_i}{\sum_i \tilde{w}_i}.$$

Training minimizes:

$$\mathcal{J}(\phi) = -\hat{V}_{\text{obj}} + \lambda_2 (\|\alpha\|_2^2 + \|\eta\|_2^2) + \lambda_{\text{var}} \sqrt{\frac{1}{t} \text{Var}(\text{contrib})},$$

where t is minibatch size and `contrib` is the per-sample term used by the selected estimator.

7 Context Reweighting

To handle context shift across rounds, the pipeline can use a target context distribution $q(\theta)$ (for example `best_round_in_k`) and per-sample ratio:

$$\omega(\theta) = \frac{q(\theta)}{p_t(\theta)}$$

estimated from histograms and clipped. These context weights are used:

- in training (`--context-match-sampling`),
- in evaluation (reported `*_q_*` metrics).

8 Model-Based IRT Value Metrics

The pipeline reports model-based values using:

$$\Pr(\text{correct} = 1 \mid \theta, \delta) = \sigma(\theta - \delta), \quad r(\theta, \delta) = \sigma(\theta - \delta)(\delta - \delta_{\min}).$$

For policy π :

$$V_{\text{IRT}}(\pi) = \mathbb{E}_{\theta \sim p} \mathbb{E}_{\delta \sim \pi(\cdot \mid \theta)} [r(\theta, \delta)],$$

estimated by Monte Carlo (`mc_policy_value`).

The oracle curve is:

$$\delta^*(\theta) \in \arg \max_{\delta \in [\delta_{\min}, \delta_{\max}]} \sigma(\theta - \delta)(\delta - \delta_{\min}),$$

and the corresponding value is reported as `optimal_irt_value`.

9 Experiment

9.1 Protocol

We ran the simplified cumulative-last-round pipeline in `offpolicy_ktm_pipeline/run_last_round_simple.py` on three datasets:

- PIX (`pix_processed.csv`),
- SkillBuilder (`skill_builder_2015_mintrace10.csv`),
- Assistments-like (`data_with_answer_number.csv`).

Common settings:

- `ktm_c=0.1`, strict 1PL KTM with no intercept;
- behavior model: truncated Gaussian with $\mu(\theta)$ degree 1 and $\log \sigma(\theta)$ degree 2;
- context target: “best round in first $N/2$ ” (closest to last-round context);
- context reweighting with histogram ratio, `context_bins=60`, `context_ratio_clip=10`;
- IS weight cap `max_weight=20`, policy sigma floor 0.2;
- rounds capped to 20 (`tail` strategy), training scope cumulative at final round;
- objectives trained separately: IPS, SNIPS, DR, MIS, DM.

Batch size followed a rule-of-thumb targeting roughly 50–100 updates per epoch: 4096 (Assistments), 8192 (SkillBuilder), 32768 (PIX).

9.2 Final-Round Learned Policy Values

Table 1 reports the final-round values of the *learned policy* under each estimator. Two robust trends are visible:

- For PIX, DR/SNIPS/MIS policies are close in value, with large ESS.
- For Assistments and SkillBuilder, IPS/MIS-optimized policies tend to give higher IPS/MIS metrics, while DR/SNIPS-optimized policies are more conservative.

This confirms that objective choice primarily changes the policy shape in regions with strong weighting effects, while remaining in a similar performance band for stable datasets.

Table 1: Final-round learned policy values across datasets/objectives (ESS in thousands).

Dataset	Obj	IPS	SNIPS	DR	MIS	DM	ESS _{log} (k)	ESS _{mix} (k)
assistments	ips	0.703	0.677	0.675	0.708	0.616	157.677	155.989
assistments	snips	0.698	0.676	0.676	0.704	0.621	157.512	155.994
assistments	dr	0.694	0.676	0.676	0.701	0.626	156.845	155.567
assistments	mis	0.706	0.679	0.678	0.712	0.616	155.006	153.381
assistments	dm	0.515	0.547	0.566	0.516	0.724	86.121	100.156
skillbuilder	ips	1.128	1.020	1.005	1.127	0.993	419.337	422.537
skillbuilder	snips	1.080	0.997	0.978	1.080	0.967	441.189	443.839
skillbuilder	dr	1.048	0.991	0.972	1.049	0.964	511.248	516.276
skillbuilder	mis	1.125	1.020	1.003	1.125	0.992	422.379	425.624
skillbuilder	dm	1.021	0.980	0.961	1.023	0.954	540.528	546.522
pix	ips	2.638	2.864	2.863	2.933	2.582	497.940	908.155
pix	snips	2.561	2.853	2.851	2.902	2.572	424.182	875.189
pix	dr	2.590	2.865	2.867	2.913	2.586	423.558	844.858
pix	mis	2.626	2.855	2.859	2.935	2.575	465.723	903.800
pix	dm	2.576	2.855	2.856	2.889	2.584	429.977	873.509

9.3 Adaptive Behavior Policy by Round

The adaptive behavior policy value was estimated at *every* round using context-reweighted estimators on cumulative seen data, and we also report context-reweighted batch reward means. Full per-round values are included in the generated tables below.

9.4 Policy Curves

For each dataset we plot:

1. Learned policies from all estimators as mean difficulty $\mathbb{E}[\delta \mid \theta]$ with the optimal IRT policy shown as dotted curve.
2. The same learned curves plus behavior policy snapshots (round 1, mid, last).

9.5 Discussion

Across datasets, the objective family mostly changes policy aggressiveness rather than completely reordering performance. In PIX, all non-DM objectives remain close with high ESS, suggesting low variance pressure after clipping. In Assistments/SkillBuilder, differences are larger, and DR/SNIPS provide more stable trade-offs against ESS than raw IPS-like objectives. Behavior-evolution plots show that learned policies generally move away from early-round behavior toward higher expected difficulty in high- θ regions, while the dotted IRT-optimal policy remains an upper reference envelope.

10 Main Outputs

Each run writes:

- `ktm_dataframe.csv`
- `round_bucket_mapping.csv`

Table 2: Assistments: behavior policy values by round (context-reweighted).

Round	BatchReward _q	IPS _q	SNIPS _q	DR _q	MIS _q	DM _q
1	0.563	0.554	0.552	0.552	0.554	0.634
2	0.579	0.562	0.550	0.544	0.556	0.631
3	0.600	0.570	0.567	0.567	0.567	0.637
4	0.574	0.565	0.550	0.543	0.559	0.641
5	0.619	0.575	0.567	0.565	0.571	0.638
6	0.599	0.576	0.569	0.569	0.573	0.641
7	0.636	0.580	0.568	0.565	0.578	0.641
8	0.613	0.584	0.579	0.578	0.583	0.643
9	0.635	0.593	0.591	0.592	0.592	0.639
10	0.631	0.592	0.589	0.588	0.591	0.644
11	0.622	0.591	0.581	0.575	0.589	0.643
12	0.631	0.591	0.581	0.579	0.590	0.646
13	0.612	0.593	0.584	0.579	0.592	0.649
14	0.610	0.591	0.584	0.582	0.591	0.651
15	0.632	0.593	0.585	0.584	0.593	0.646
16	0.612	0.592	0.583	0.582	0.592	0.649
17	0.615	0.593	0.583	0.578	0.592	0.650
18	0.632	0.591	0.582	0.579	0.590	0.650
19	0.625	0.594	0.579	0.573	0.592	0.648
20	0.625	0.608	0.603	0.604	0.608	0.652

Table 3: SkillBuilder: behavior policy values by round (context-reweighted).

Round	BatchReward _q	IPS _q	SNIPS _q	DR _q	MIS _q	DM _q
1	0.656	0.570	0.568	0.583	0.570	0.704
2	0.763	0.609	0.606	0.621	0.608	0.699
3	0.806	0.633	0.630	0.645	0.633	0.697
4	0.803	0.652	0.653	0.664	0.651	0.708
5	0.830	0.659	0.659	0.671	0.659	0.709
6	0.842	0.665	0.664	0.677	0.665	0.708
7	0.809	0.670	0.665	0.681	0.670	0.711
8	0.819	0.672	0.666	0.683	0.672	0.711
9	0.825	0.673	0.665	0.685	0.673	0.709
10	0.822	0.680	0.673	0.693	0.680	0.715
11	0.817	0.682	0.674	0.694	0.681	0.714
12	0.825	0.686	0.677	0.698	0.685	0.716
13	0.828	0.689	0.682	0.701	0.689	0.718
14	0.824	0.693	0.685	0.704	0.692	0.721
15	0.827	0.693	0.686	0.704	0.692	0.720
16	0.835	0.695	0.689	0.707	0.695	0.722
17	0.831	0.696	0.690	0.708	0.696	0.723
18	0.846	0.697	0.691	0.709	0.697	0.724
19	0.846	0.699	0.694	0.711	0.699	0.724
20	0.902	0.741	0.737	0.756	0.744	0.753

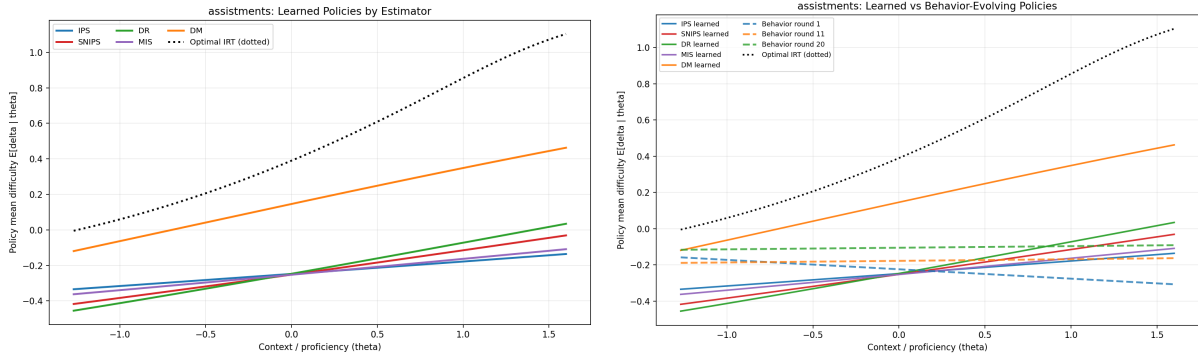


Figure 1: Assistments policy curves: learned estimators vs optimal IRT, and behavior evolution.

Table 4: PIX: behavior policy values by round (context-reweighted).

Round	BatchReward _q	IPS _q	SNIPS _q	DR _q	MIS _q	DM _q
1	0.325	3.347	1.726	1.555	3.347	1.357
2	1.074	1.855	1.309	0.902	0.777	0.806
3	1.726	0.655	0.581	0.602	0.413	0.595
4	2.009	0.763	0.746	0.719	0.492	0.709
5	2.526	1.527	1.346	1.206	1.019	1.164
6	2.683	1.873	1.717	1.666	1.667	1.602
7	2.666	1.336	1.514	1.366	1.234	1.343
8	2.757	1.833	1.896	1.853	1.909	1.787
9	2.638	2.285	2.401	2.505	2.526	2.364
10	2.608	2.117	2.719	2.764	2.792	2.590
11	2.358	1.646	2.646	2.567	2.706	2.444
12	2.349	1.753	2.668	2.606	2.709	2.473
13	2.182	1.663	2.584	2.500	2.558	2.390
14	1.789	1.066	2.025	1.686	1.665	1.727
15	1.808	1.101	1.991	1.725	1.701	1.773
16	2.353	2.368	2.703	2.691	2.702	2.483
17	1.947	1.650	2.234	2.091	2.095	2.046
18	2.038	1.696	2.251	2.125	2.131	2.070
19	2.081	1.768	2.282	2.169	2.177	2.102
20	2.074	1.985	2.281	2.248	2.249	2.205

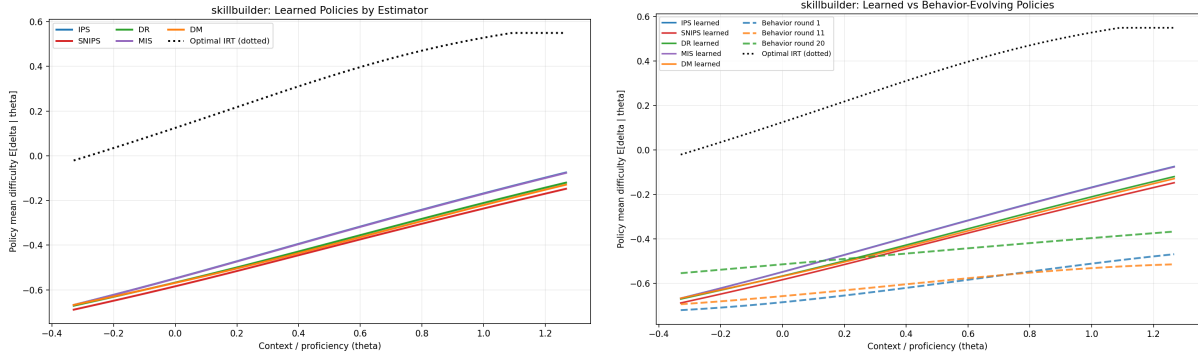


Figure 2: SkillBuilder policy curves: learned estimators vs optimal IRT, and behavior evolution.

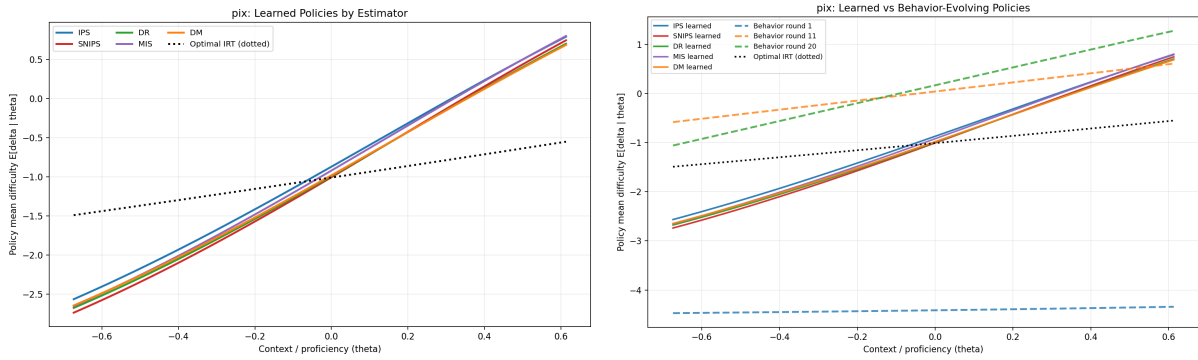


Figure 3: PIX policy curves: learned estimators vs optimal IRT, and behavior evolution.

- gaussian_fit_by_order.csv
- offpolicy_tuples.csv
- policy_round_results.csv
- policy_value_table.csv
- policy_coefficients_by_round.csv
- optional plots/animation.

11 Reference Command

```
python offpolicy_ktm_pipeline/run_full_pipeline.py \
  --input-csv pix_mapping/pix_processed.csv \
  --objective ips \
  --denominator mixture \
  --train-scope cumulative \
  --context-target best_round_in_k \
  --context-k-rounds 5 \
  --balanced-minibatch \
  --context-match-sampling \
  --sigma-floor 0.2
```